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Personalized Adaptive Learning Management System Using Meta-Reinforcement Learning and Large Language Models

A CAPSTONE PROJECT REPORT

*A Capstone Project Report submitted in partial fulfilment of the requirements for
the Course of*

MLA0301-REINFORCEMENT LEARNING

to the award of the degree of
BACHELOR OF ENGINEERING

IN

ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

Submitted by

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BONAFIDE CERTIFICATE

This is to certify that the Capstone Project entitled “**Personalized Adaptive Learning Management System Using Meta-Reinforcement Learning and Large Language Models**” has been carried out by **Yoga Madha A** under the supervision of **Dr.G.A.Senthil** and is submitted in partial fulfilment of the requirements for the current semester of the **B-TECH Artificial Intelligence and Machine Learning** program at Saveetha Institute of Medical and Technical Sciences, Chennai.

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DECLARATION

I, **YogaMadha(192425058)** SIMATS Engineering, Saveetha Institute of Medical and Technical Sciences, Chennai, hereby declare that the Capstone Project Work entitled “**Personalized Adaptive Learning Management System Using Meta-Reinforcement Learning and Large Language Models**” is the result of our own bonafide efforts. To the best of our knowledge, the work presented herein is original, accurate, and has been carried out in accordance with the principles of engineering ethics and academic integrity. All sources, references, data, and other materials used in the project have been appropriately acknowledged. We further declare that the data, results, analysis, and findings presented in this report have not been fabricated, falsified, or manipulated. Any external tools, software, datasets, computational resources, or AI-assisted tools used during the project have been appropriately disclosed. We confirm that all applicable ethical, academic, institutional, and professional requirements have been duly followed throughout the planning, execution, analysis, and documentation of the project.

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Individual Contribution Statement

Student Name / Register No.	Specific Responsibilities	Design & Development Contribution	Testing & Analysis Contribution	Report Contribution	Approx. Contribution (%)
Chandana M (192472250)	Module 1 – Personalized Learning Engine	Student profile creation, learning behavior analysis, Meta-Reinforcement Learning, adaptive learning path and content recommendation	Tested learning behavior analysis, adaptive learning path and recommendation process	Prepared Module 1 documentation	33.33%
Yoga Madha A (192425058)	Module 2 – AI Tutor Using Large Language Models	Developed personalized explanations, AI-based doubt solving, adaptive quiz and assignment generation	Tested AI responses, quiz generation, assignments and instant feedback	Prepared Module 2 documentation	33.34%
KushithaG (192425329)	Module 3 – Performance Analytics & Recommendation Dashboard	Developed progress tracking, performance visualization, learning analytics and personalized recommendations	Tested dashboard, analytics, learning trends and recommendation outputs	Prepared Module 3 documentation	33.33%
Total	All Modules	Complete system	Complete system testing	Complete project report	100%

ABSTRACT

The development of effective personalized learning systems is an important engineering challenge because conventional learning platforms generally provide similar learning sequences and content to students with different knowledge levels, learning speeds, and performance patterns. This limitation can result in ineffective learning paths, insufficient attention to weak areas, and delayed feedback. There is therefore a need for an intelligent learning management system that can continuously understand learner behavior and provide personalized learning support.

This project proposes a **Personalized Adaptive Learning Management System Using Meta-Reinforcement Learning and Large Language Models**. The system is designed as an integrated platform consisting of three core modules: the **Personalized Learning Engine**, **AI Tutor Using Large Language Models**, and **Performance Analytics & Recommendation Dashboard**. The Personalized Learning Engine analyzes student profiles, learning activities, quiz performance, mistakes, completion patterns, and learning consistency to determine suitable learning actions. Meta-Reinforcement Learning is used to adapt the learning strategy according to the changing state of each student.

The AI Tutor provides personalized explanations, interactive doubt solving, adaptive quizzes, assignments, and instant feedback based on the learner's knowledge level and previous performance. The Performance Analytics and Recommendation Dashboard tracks progress, visualizes performance, identifies weak and strong areas, detects learning gaps, and generates improvement recommendations. Data from student interactions are continuously returned to the learning engine, creating an adaptive learning cycle.

The major expected result is a learning platform capable of dynamically selecting suitable learning activities instead of following a fixed learning sequence. The system can provide targeted practice for weak areas, increase difficulty when mastery is demonstrated, and provide additional support when performance decreases. Analytics also provide meaningful insights into learning progress and recommendation effectiveness. The principal conclusion is that integrating **Meta-Reinforcement Learning and Large Language Models** can provide a more adaptive, interactive, and personalized learning experience. The proposed system establishes a continuous feedback loop in which student behavior, AI tutoring, performance analysis, and learning recommendations work together to progressively improve the learning strategy.

Keywords: Personalized Learning, Meta-Reinforcement Learning, Large Language Models, AI Tutor, Learning Analytics, Adaptive Recommendation

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CHAPTER 1

INTRODUCTION AND ENGINEERING PROBLEM

CHAPTER 1 — INTRODUCTION AND ENGINEERING PROBLEM

1.1 Background

The rapid growth of digital education has created a need for learning platforms that can respond to the individual requirements of students. Traditional Learning Management Systems generally provide predefined courses, fixed learning sequences, and common assessment patterns. However, students differ in their knowledge levels, learning pace, previous performance, strengths, weaknesses, and learning behavior. A fixed learning approach may therefore provide the same content even when students require different levels of explanation or practice.

The proposed project, **Personalized Adaptive Learning Management System Using Meta-Reinforcement Learning and Large Language Models**, addresses this limitation by combining adaptive learning techniques with Artificial Intelligence. The system is organized into three core modules: **Personalized Learning Engine**, **AI Tutor Using Large Language Models**, and **Performance Analytics & Recommendation Dashboard**. These modules operate together as a continuous learning cycle rather than as independent components.

The Personalized Learning Engine uses student information and learning behavior as the basis for selecting appropriate learning actions. It considers factors such as quiz scores, incorrect answers, attempts, time spent learning, topic completion, repeated mistakes, and improvement trends. Meta-Reinforcement Learning is applied to continuously adapt the learning strategy according to the changing learning state of the student.

The AI Tutor uses Large Language Models to provide personalized explanations, answer student questions, generate adaptive quizzes and assignments, and provide immediate feedback. The analytics module processes learning data to visualize performance, identify weak areas, detect learning gaps, and generate recommendations. The updated learning information is then returned to the learning engine to improve subsequent recommendations.

Thus, the project provides an integrated engineering solution in which **student behavior** →

adaptive decision-making → AI tutoring → performance analysis → improved learning recommendation forms a continuous feedback loop.

1.2 Need for the Project

Why the Problem Needs to Be Addressed

Conventional learning platforms often follow a common learning sequence for different students. Such systems may not adequately respond when a student struggles with a particular topic or progresses faster than expected. Students may also require different explanation styles, difficulty levels, practice activities, and feedback mechanisms.

An adaptive system is therefore required to continuously analyze learning behavior and select suitable learning activities based on the student's current learning state.

Who Is Affected by the Problem

The problem mainly affects:

- **Students**, who may receive content that is too easy, too difficult, or poorly aligned with their learning needs.
- **Teachers and faculty**, who may find it difficult to continuously monitor individual student performance.
- **Educational institutions**, which require effective tools for monitoring learning progress and identifying students who need additional support.

Existing Limitations

Existing conventional learning platforms may have limitations such as:

- Fixed learning sequences.
- Limited personalization.
- Common difficulty levels for different learners.
- Delayed or generalized feedback.

- Limited identification of repeated mistakes and learning gaps.
- Manual monitoring of individual student progress.
- Limited integration between tutoring, assessment, analytics, and recommendations.

Engineering Significance

The project has engineering significance because it combines **Meta-Reinforcement Learning, Large Language Models, learning analytics, adaptive recommendations, and interactive software components** into a single system. The engineering challenge is to maintain continuous information flow between these components while producing appropriate learning decisions for individual students.

1.3 Problem Statement

The engineering problem is to **design and develop an intelligent adaptive learning management system that can continuously analyze individual student behavior, dynamically select appropriate learning activities, provide context-aware AI tutoring, and generate performance-based recommendations.**

The problem involves multiple interacting factors, including student knowledge level, learning goals, quiz performance, mistakes, learning time, topic completion, learning consistency, and previous learning history. These factors can change continuously during the learning process.

The system must also address conflicting requirements. It should provide sufficient practice to students who require support while allowing students who demonstrate mastery to progress faster. It must provide personalized content without making the learning process unnecessarily complex. At the same time, the system must maintain continuous communication between the adaptive learning engine, AI tutor, and analytics components.

The engineering solution must operate within realistic constraints such as available learning data, computational resources, model response time, system scalability, data consistency, and reliable integration between different software components.

1.4 Project Aim

The aim of this project is to **design and develop a Personalized Adaptive Learning**

Management System that uses Meta-Reinforcement Learning and Large Language Models to analyze student learning behavior, personalize learning paths, provide intelligent tutoring, and generate continuous performance-based recommendations.

1.5 Project Objectives

1. **To design** a personalized learning management architecture consisting of three integrated core modules.
2. **To develop** a Personalized Learning Engine for maintaining student profiles and analyzing learning behavior.
3. **To model** student learning states, possible learning actions, and reward-based adaptation using Meta-Reinforcement Learning.
4. **To implement** an adaptive learning path that dynamically recommends suitable topics, difficulty levels, practice activities, and learning resources.
5. **To develop** an AI Tutor using Large Language Models for personalized explanations, doubt solving, quiz generation, assignment generation, and instant feedback.
6. **To implement** a Performance Analytics and Recommendation Dashboard for tracking progress, topic mastery, learning trends, and weak areas.
7. **To integrate** student responses and performance data across all three modules to establish a continuous adaptive learning loop.
8. **To test** the system using different student learning conditions, performance levels, and learning activities.
9. **To evaluate** the effectiveness of adaptive recommendations, AI tutoring, performance analysis, and overall system integration.

1.6 Scope

What Is Included

The project includes:

- Student profile creation and management.
- Learning behavior monitoring.
- Student performance and activity tracking.
- Meta-Reinforcement Learning-based adaptive decision-making.
- Personalized learning paths.
- Dynamic learning activity recommendations.
- Large Language Model-based AI tutoring.
- Personalized explanations and doubt solving.
- Adaptive quizzes and assignments.
- Instant feedback.
- Performance analytics and visualization.
- Weak-area and learning-gap identification.
- Personalized improvement recommendations.
- Adaptive learning insights.
- Faculty/Admin monitoring dashboard.
- Integration of the three modules into a continuous learning cycle.

What Is Excluded

The project does not aim to replace teachers or provide a complete institutional education management system. It does not cover areas such as admission management, fee management, examination administration, or complete university administration.

The system is focused specifically on **personalized learning, AI tutoring, learning analytics, and adaptive recommendations.**

Assumptions

The project assumes that:

- Students provide sufficient profile and learning activity information.
- Quiz and assignment responses are available for performance analysis.
- Learning data can be stored and processed by the system.
- The AI model can generate appropriate educational responses.
- The adaptive engine receives updated learning information after student interactions.

Boundary Conditions

The system operates within the defined three-module architecture. Its recommendations depend on the quality and availability of student learning data. The accuracy of analytics and personalization can therefore vary according to the amount and quality of available learning information.

1.7 Expected Engineering Outcome

The expected engineering outcome is a **working prototype of a Personalized Adaptive Learning Management System** integrating Meta-Reinforcement Learning and Large Language Models.

The final system is expected to provide:

- A **Personalized Learning Engine** for analyzing learner behavior and adapting learning paths.
- An **AI Tutor** for personalized explanations, doubt solving, assessments, and feedback.
- A **Performance Analytics & Recommendation Dashboard** for visualizing learning progress and generating improvement recommendations.
- A continuous data flow between the three modules.
- Dynamic recommendations based on the student's current learning state.
- Identification of weak areas, learning gaps, and improvement patterns.

CHAPTER 2

LITERATURE REVIEW AND EXISTING SOLUTIONS

2.1 Review Method

A structured review of existing research and technologies was carried out to understand the current development of personalized and adaptive learning systems. Relevant literature was identified using academic sources and digital research databases, with particular attention given to **Artificial Intelligence in Education, Adaptive Learning, Intelligent Tutoring Systems, Reinforcement Learning, Meta-Reinforcement Learning, Large Language Models, and Learning Analytics**.

Search terms such as *personalized learning*, *adaptive learning systems*, *reinforcement learning in education*, *intelligent tutoring systems*, *large language models in education*, and *AI-based learning analytics* were considered. Recent systematic reviews and research studies were given greater importance to understand current developments and unresolved engineering challenges.

A 2025 systematic literature review examined 89 studies on Reinforcement Learning in education and reported applications including personalized interventions, adaptive difficulty, content scheduling, and instructional decision-making. It also identified the need for larger real-user evaluations and stronger methodological approaches. (ScienceDirect)

Recent literature on LLMs in education was also reviewed. A 2026 systematic review examined 88 empirical studies and found that Intelligent Tutoring Systems were among the most common applications of LLMs in education, while also highlighting concerns related to reliability, privacy, fairness, and over-reliance. (ScienceDirect)

2.2 Review of Existing Work

2.2.1 Adaptive Learning Systems

Adaptive learning systems attempt to modify educational content according to individual learner characteristics and performance. Earlier adaptive education research distinguishes between Intelligent Tutoring Systems and adaptive hypermedia systems and emphasizes the importance of adapting instructional content to learners. (ScienceDirect)

More recent research has expanded adaptive learning through Machine Learning, Deep Learning, Reinforcement Learning, and learner analytics. A 2025 systematic review of AI-powered adaptive education analyzed 142 empirical studies and reported the use of AI techniques for personalized and

data-driven learning experiences. (Springer)

However, many adaptive systems remain dependent on predefined rules, learner models, or limited performance indicators. They may adapt content effectively but may not continuously optimize a personalized learning policy based on changing student behavior.

2.2.2 Reinforcement Learning in Education

Reinforcement Learning provides a suitable framework for adaptive learning because it allows an agent to select actions based on the current state and receive rewards according to the outcome. In education, possible actions can include changing content difficulty, selecting learning resources, providing feedback, or deciding the next learning activity.

A systematic review published in 2025 examined RL applications in education and found applications in areas such as learning outcomes, student engagement, adaptive difficulty, scheduling, and personalized feedback. However, the review identified challenges including methodological limitations and insufficient large-scale evaluation with real learners. (Springer)

Another 2024 scoping review highlighted the growing use of RL in education and its potential for interactive and adaptive learning environments. It also emphasized the importance of considering pedagogical factors and possible bias when applying RL to education. (ScienceDirect)

These findings support the use of reinforcement-based decision-making in the proposed Personalized Learning Engine.

2.2.3 Intelligent Tutoring Systems

Intelligent Tutoring Systems are designed to provide individualized instructional support using computational models of learners and educational content. Traditional ITS approaches commonly use predefined knowledge models, rules, or structured instructional strategies.

Recent ITS research has moved toward AI-driven systems capable of providing more flexible interaction. A 2025 IEEE review reported a shift toward systems using Large Language Models and adaptive learning architectures to provide more personalized and human-like learning support. (IEEE Xplore)

The limitation is that an AI tutor operating independently may provide useful explanations but may not maintain a complete adaptive learning strategy across the student's long-term learning journey.

2.2.4 Large Language Models as AI Tutors

Large Language Models have introduced new possibilities for educational tutoring. They can generate explanations, answer questions, create educational material, evaluate responses, and provide conversational assistance.

A review of LLM-based virtual tutors identified educational applications including assessment and learning assistance and examined how students and teachers are involved in LLM-based educational solutions. (Springer)

A recent systematic review of 88 empirical studies found that LLMs can support academic performance, engagement, accessibility, assessment, and intelligent tutoring. At the same time, it identified concerns involving technical reliability, privacy, fairness, and student over-reliance on AI. (ScienceDirect)

Therefore, LLMs are highly suitable for the proposed AI Tutor, but they need to be integrated with learner-state information and performance analytics rather than being used as an isolated chatbot.

2.2.5 Learning Analytics and Recommendation Systems

Learning analytics systems collect and process student activity data to identify learning patterns, performance trends, weak areas, and learning gaps. These systems can support students and instructors by transforming raw activity data into meaningful insights.

Recent research indicates that AI-supported personalized learning increasingly combines learner modelling, adaptive systems, analytics, and intelligent tutoring. A 2026 review of personalized learning research identified personalized learning, adaptive learning, and intelligent tutoring as major areas of research and highlighted continuing development of AI-enabled educational systems. (Springer)

However, analytics systems are often used mainly for visualization and monitoring rather than being directly connected to a learning policy that changes the next activity.

2.2.6 LLMs and Adaptive Learning Path Optimization

Recent research has also explored combining LLMs with knowledge structures for adaptive learning-path optimization. A 2026 study investigated the combination of LLMs and knowledge graphs for personalized learning paths in Intelligent Tutoring Systems. (eJournal of ePublishing)

This demonstrates a movement toward integrated intelligent learning systems. However, there remains an opportunity to combine **continuous learner-state adaptation, reinforcement-based decision-making, LLM tutoring, and performance analytics** within a single application architecture.

2.3 Comparative Analysis

Existing Approach	Advantages	Limitations	Relevance to Proposed Work
Traditional Learning Management System	Easy to implement; structured course delivery; centralized content	Fixed learning sequence; limited personalization; common content for different students	Provides the basic platform structure that the proposed system improves through adaptive learning
Rule-Based Adaptive Learning	Simple decision-making; predictable behavior; easy to understand	Depends on predefined rules; limited ability to adapt to changing learner behavior	Motivates the use of learning policies that can continuously adapt
Intelligent Tutoring System	Provides individualized guidance and feedback	May depend on predefined knowledge models and limited interaction patterns	Forms the conceptual basis for the proposed AI Tutor
Reinforcement Learning-Based Education	Supports adaptive decisions and personalized interventions	Requires appropriate reward design; evaluation with real users remains challenging	Used for dynamic learning-action selection in Module 1
LLM-Based AI Tutor	Natural conversation; explanations; question answering; content generation	Reliability, privacy, fairness and over-reliance concerns	Used for personalized explanations, doubt solving, quizzes and feedback
Learning Analytics Dashboard	Provides visual performance information and identifies learning patterns	Often focuses on monitoring rather than automatically changing learning activities	Used as the basis for Module 3 and connected to recommendations
LLM + Knowledge-Based Adaptive Systems	Supports personalized learning paths and flexible content generation	Integration complexity; requires reliable learner modelling and structured information	Demonstrates the potential of combining AI technologies for adaptive learning
Proposed System	Combines adaptive learning, Meta-RL, LLM tutoring, analytics and recommendations in one continuous loop	Requires integration of multiple AI components and reliable learner data	Provides the complete integrated solution proposed in this project

The comparison shows that individual technologies address specific parts of personalized education, while the proposed system focuses on **integrating these capabilities into one continuous adaptive**

learning cycle.

2.4 Research / Engineering Gap

The literature indicates that significant progress has been made in adaptive learning, Reinforcement Learning, Intelligent Tutoring Systems, LLM-based education, and learning analytics. However, several engineering gaps remain.

1. Limited Integration of AI Components

Many existing approaches focus on one major capability, such as adaptive sequencing, AI tutoring, or analytics. There is a need for stronger integration between these components.

2. Limited Continuous Adaptation

Some adaptive systems use predefined rules or fixed learner models. A more dynamic approach is required where the learning strategy changes continuously according to new student interactions and performance.

3. Separation Between Tutoring and Learning Policy

An LLM tutor can answer questions and provide explanations, but tutoring interactions do not always directly update the student's long-term learning strategy. The proposed system addresses this by returning student interaction data to the Personalized Learning Engine.

4. Analytics Not Directly Connected to Learning Decisions

Many analytics dashboards primarily visualize performance. The unresolved engineering challenge is to use those analytics directly to determine what the student should learn next.

5. Need for Continuous Feedback

The learning process requires a feedback loop in which student activity influences the learning state, recommendations, tutoring, and subsequent learning decisions. This type of integrated cycle is central to the proposed system.

6. Practical AI Integration Challenges

Current research also identifies concerns involving AI reliability, privacy, fairness, explainability, and over-reliance. (ScienceDirect) These factors must be considered when developing a practical

educational AI system.

2.5 Proposed Contribution

The proposed project addresses the identified gap by developing a **single Personalized Adaptive Learning Management System** that integrates three complementary modules.

Module 1 — Personalized Learning Engine

The first module maintains student profiles, analyzes learning behavior, models the learner's current state, and uses Meta-Reinforcement Learning to select appropriate learning actions. The learning path can therefore change according to student performance and progress.

Module 2 — AI Tutor Using Large Language Models

The second module provides personalized explanations, doubt solving, adaptive quizzes, assignments, and instant feedback. The tutor uses information from the learner's current state to provide more relevant assistance.

Module 3 — Performance Analytics & Recommendation Dashboard

The third module converts student activity into meaningful performance insights. It identifies weak areas, learning gaps, performance trends, and improvement opportunities and generates recommendations connected to the adaptive learning engine.

Overall Contribution

The main contribution is the **integration of these three modules into a continuous adaptive learning loop**:

Student Activity → Learning Behavior Analysis → Meta-Reinforcement Learning → Personalized Recommendation → AI Tutor / Assessment → Student Response → Performance Analysis → Updated Learning State → Improved Recommendation

This approach differs from systems that treat tutoring, analytics, and adaptive learning as separate functions. The proposed architecture allows information generated in one module to influence the behavior of the other modules, creating an increasingly personalized learning experience. This integrated direction is consistent with recent research calling for stronger connections between AI technologies, learner modelling, adaptivity, and educational analytics. (academic-publishing.org)

CHAPTER 3

ENGINEERING DESIGN & TRADE-OFFS, SUSTAINABILITY, ETHICS, SAFETY, RISK & SECURITY

3.1 Requirements

3.1.1 Stakeholder / User Requirements

Stakeholder / User	Requirement
Student	Create and manage a personal learning profile containing subjects, topics, goals, knowledge level and learning history.
Student	Receive a learning path that adapts to individual performance and learning behavior.
Student	Receive personalized explanations, doubt-solving support, quizzes, assignments and instant feedback.
Student	View progress, topic mastery, weak areas and improvement recommendations.
Faculty	Monitor individual student progress and identify students requiring additional academic support.
Faculty/Admin	View subject-level, topic-level and overall learning trends.
System Administrator	Maintain reliable operation, user access and protection of stored learning information.
Project Team	Maintain integration between the three core modules and ensure that student interactions update the adaptive learning cycle.

3.1.2 Functional & Non-Functional Requirements

Requirement Type	Measurable Target
Functional	System shall support creation and management of individual student profiles.
Functional	System shall record quiz scores, attempts, incorrect answers, learning time and topic completion.
Functional	System shall determine the student's current learning state from available performance data.
Functional	System shall recommend an appropriate next learning activity based on the current learning state.
Functional	System shall provide AI-generated explanations and doubt-solving assistance.
Functional	System shall generate adaptive quizzes and assignments.
Functional	System shall provide immediate feedback after student responses.
Functional	System shall display performance, topic mastery and learning trends.
Functional	System shall identify weak areas and generate improvement

Requirement Type	Measurable Target
	recommendations.
Functional	System shall provide a faculty/admin monitoring view.
Functional	System shall return new learning data to the adaptive learning engine.
Non-Functional	Common dashboard interactions should respond within approximately 3 seconds under normal prototype conditions.
Non-Functional	The system shall maintain consistent student data across all three modules.
Non-Functional	The interface shall be usable through a standard modern web browser.
Non-Functional	User access shall be controlled according to the defined user role.
Non-Functional	The architecture shall allow individual modules to be modified without redesigning the complete system.

3.2 Constraints & Applicable Standards

Only standards directly relevant to the proposed software system and its handling of AI-generated educational information are considered.

Constraints

Constraint Category	Applicable Constraint
Technical	The quality of adaptive recommendations depends on the availability and quality of student learning data.
Technical	LLM responses can vary and may occasionally produce inaccurate or unsuitable educational content.
Technical	Meta-Reinforcement Learning requires appropriate state, action and reward definitions.
Economic	The prototype should use commonly available software and computational resources to keep development cost low.
Ethical	Student learning data should not be exposed unnecessarily, and AI recommendations should not unfairly disadvantage learners.
Security / Regulatory	Personal and educational information must be protected against unauthorized access and inappropriate disclosure.

Applicable Standards / Codes

Standard / Code	Requirement	How Applied in This Project
ISO/IEC 25010:2023	Defines a software product quality model covering characteristics such as performance efficiency, reliability, security, maintainability and usability.	Used as a reference when defining non-functional requirements and evaluating software quality.

Standard / Code	Requirement	How Applied in This Project
ISO/IEC 27001:2022	Specifies requirements for establishing and maintaining an Information Security Management System.	Used as a security-management reference for protecting student information and controlling access.
ISO/IEC 23894:2023	Provides guidance for managing risks associated with AI systems.	Used as a reference for identifying risks related to AI recommendations and LLM-generated educational responses.
NIST AI RMF 1.0	Provides a framework for managing AI risks and promoting trustworthy AI development and use.	Used as a reference for considering reliability, transparency, fairness and risk management in AI-based functions.

3.3 Design Specifications

Parameter	Target / Requirement	Unit	Priority
Student Profile	Store learner-specific academic and learning information	Profile	High
Learning State	Updated from recent student activity and performance	State	High
Learning Actions	Support new topic, revision, difficulty change, practice and resource recommendation	Actions	High
Recommendation Update	Update after meaningful learning interaction	Event	High
Quiz Generation	Generate questions according to topic and learner level	Quiz	High
AI Response	Context-aware educational response	Response	High
Performance Tracking	Record quiz, assignment and topic performance	Data	High
Topic Mastery	Display topic-level learning status	% / Level	High
Dashboard	Interactive performance and recommendation views	Interface	Medium
Role Access	Student and faculty/admin views	Role	High
Response Time	Normal dashboard operation approximately ≤ 3 seconds	Seconds	Medium
Data Consistency	Same learner state reflected across all modules	Requirement	High

3.4 Alternative Solutions & Evaluation

Alternative 1: Rule-Based Adaptive Learning

A rule-based system can select learning activities using predefined conditions such as quiz score, topic mastery and number of incorrect answers. It is simple to implement and easy to interpret but has limited ability to learn an evolving personalized policy.

Alternative 2: Conventional Machine Learning Recommendation

A supervised or recommendation-based approach can predict suitable learning resources from historical student data. It can provide useful predictions but requires sufficient training data and may not naturally model the sequential decision-making process of learning.

Alternative 3: Meta-Reinforcement Learning

Meta-Reinforcement Learning treats each learner as an adaptive environment and selects actions according to the learner's changing state and reward. It is more suitable for continuous personalization because the learning policy can be updated from student interactions.

Evaluation Matrix

Scoring: 1 = Poor, 3 = Moderate, 5 = Excellent

Criterion	Weight	Alternative 1: Rule-Based	Alternative 2: Conventional ML	Alternative 3: Meta-RL
Performance / Adaptability	30%	3	4	5
Cost	20%	5	4	3
Safety / Predictability	15%	5	4	3
Sustainability / Resource Efficiency	10%	5	4	3
Reliability	25%	5	4	4
Weighted Score	100%	4.50	4.00	3.85

The matrix indicates that **Meta-RL provides the strongest adaptive capability**, while rule-based learning provides the highest simplicity, predictability and cost efficiency. The proposed design therefore uses Meta-RL for the core adaptive decision-making while maintaining

controlled system boundaries around AI-generated recommendations.

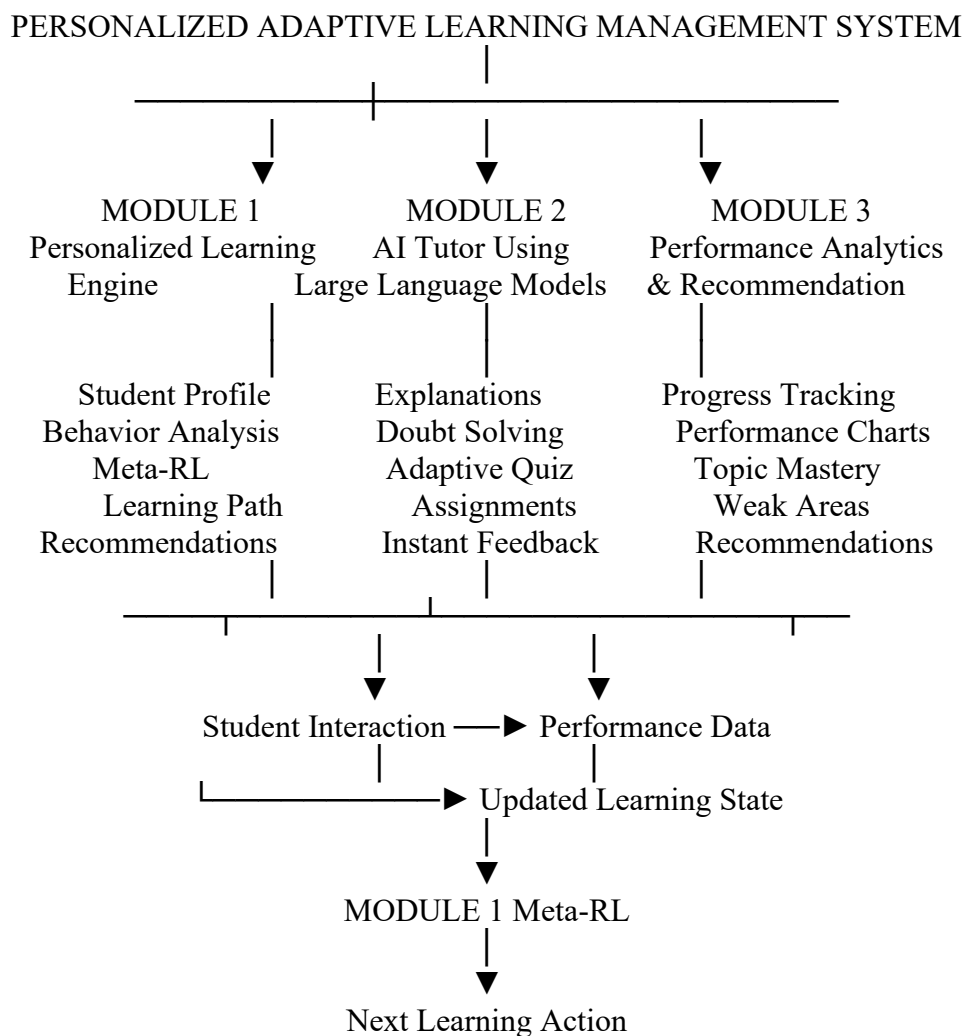
3.5 Selected Design & Detailed Design

Justification for Final Design Selection

The final design uses **Meta-Reinforcement Learning as the adaptive decision-making mechanism**, supported by an LLM-based AI Tutor and a performance analytics dashboard. Although a rule-based approach provides lower implementation complexity, Meta-RL better matches the project's requirement for continuous adaptation to changing student behavior.

The final architecture is divided into three clearly defined modules while maintaining a common learning-state and data flow.

System Architecture



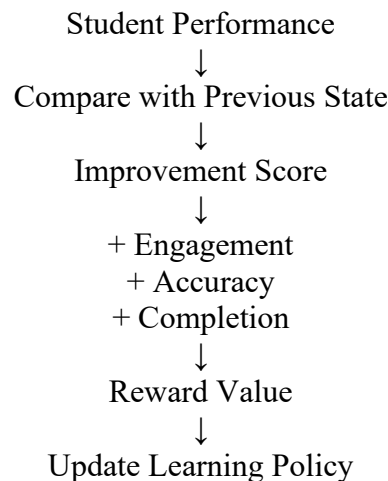
Essential Engineering Calculation — Recommendation Reward

A normalized reward can be used to represent learning improvement:

$$\text{Reward} = \text{Improvement} + \text{Engagement} + \text{Accuracy} + \text{Completion} - \text{Penalty}$$

For example, the system can normalize each factor to a value between **0 and 1**. The resulting reward provides feedback to the adaptive learning policy.

A simplified representation is:



The exact weighting of reward components can be tuned during implementation and evaluation.

Systems Approach

Module 1 maintains the learner state and determines the next suitable learning action. Module 2 delivers the required educational interaction and generates new learning data from student responses. Module 3 converts this data into performance insights and recommendations. The updated information is returned to Module 1, allowing the adaptive learning policy to continuously improve.

3.6 Trade-Off Analysis

Design Decision	Alternative Considered	Benefit	Disadvantage	Final Decision & Justification
Adaptive decision-making	Rule-based approach	Simple and predictable	Limited continuous adaptation	Meta-RL selected because continuous learner adaptation is a primary project requirement.

Design Decision	Alternative Considered	Benefit	Disadvantage	Final Decision & Justification
AI tutoring	Fixed educational content	Predictable content	Cannot provide flexible personalized interaction	LLM Tutor selected for explanations, doubt solving and dynamic learning assistance.
Quiz generation	Fixed question bank	Easy to control	Limited personalization	Adaptive generation selected to target learner level and mistakes.
Analytics	Basic numerical reports	Simple implementation	Limited visual understanding	Interactive analytics selected for easier identification of trends and weak areas.
Recommendations	Independent recommendation module	Easy separation	Does not influence learning policy	Integrated recommendations selected so analytics directly support Module 1.
System architecture	Three independent applications	Easier individual development	Data becomes disconnected	Integrated three-module architecture selected to maintain the continuous learning cycle.
AI response control	Unrestricted LLM output	Maximum flexibility	Higher risk of inaccurate responses	Context-constrained educational interaction selected to improve reliability and relevance.

3.7 Sustainability, Ethics, Safety, Risk & Security

3.7.1 Sustainability, Ethics, Safety and Security Evaluation

Domain	Consideration	Evidence Evaluated	Impact on Final Decision
Sustainability	Computational and storage requirements	Use of modular architecture, lightweight learner-state data and on-demand AI interaction	Avoid unnecessary processing and allow components to scale independently.
Ethics	Fairness of personalized recommendations	Student performance and learning behavior are used instead of unrelated personal characteristics	Recommendations are based on educational evidence rather than irrelevant attributes.
Ethics	AI-generated content reliability	Possibility of incorrect or misleading LLM responses	AI is positioned as a learning assistant rather than an unquestionable authority.
Health & Safety	Student wellbeing and responsible AI use	Risk of excessive dependence on automated recommendations	The system supports learning decisions but does not replace teachers or institutional academic judgment.
Security	Protection of student learning information	Student profiles contain personal and academic information	Role-based access and controlled information exposure are included in the design.
Security	Unauthorized access	Need to protect student and faculty/admin functions	Authentication and authorization are treated as core system requirements.
Risk	Adaptive recommendation errors	Incorrect state estimation can lead to unsuitable activities	Recommendations are evaluated using subsequent performance and updated through the feedback loop.

3.7.2 Risk Register

Risk Level: Low / Medium / High

Risk	Likelihood	Severity	Risk Level	Mitigation	Residual Risk
Incorrect AI-generated explanation	Medium	High	High	Use contextual prompts, response validation and encourage verification for important academic information	Medium
Incorrect adaptive recommendation	Medium	High	High	Evaluate recommendation outcomes and update the learning policy using feedback	Medium
Unauthorized access to student data	Low	High	High	Authentication, authorization and controlled access to student information	Low
Data inconsistency between modules	Medium	High	High	Use a common learner-state representation and validate data flow between modules	Low–Medium
AI service unavailability	Medium	Medium	Medium	Provide controlled fallback behavior and retain available learning information	Low–Medium
Excessive dependence on AI tutoring	Medium	Medium	Medium	Present AI as learning assistance and encourage active student participation	Low
Model bias in recommendations	Low–Medium	High	Medium – High	Monitor recommendation patterns and evaluate performance across different learner profiles	Medium
High computational usage	Medium	Medium	Medium	Use modular processing and invoke computationally expensive AI functions only when required	Low

CHAPTER 4

IMPLEMENTATION, TESTING & RESULTS, PROJECT MANAGEMENT

4.1 Development Approach & Resources

The project was developed using a **modular and iterative software development approach** in which the three core modules were implemented and integrated progressively. Module 1 was developed to maintain student profiles, analyze learning behavior, and generate adaptive learning decisions using Meta-Reinforcement Learning. Module 2 was integrated with the learning state to provide LLM-based tutoring, adaptive assessments, explanations, and feedback. Module 3 was then developed to analyze performance, visualize learning progress, and return recommendation information to the learning engine. Testing was performed throughout development to verify individual modules as well as the complete adaptive learning cycle.

Materials / Components / Dataset / Resources

Resource	Purpose
Student Profile Data	Stores learner characteristics, goals, knowledge level and learning history
Quiz & Assessment Data	Used for performance and learning behavior analysis
Learning Activity Data	Used to track attempts, time, completion and interaction patterns
Meta-Reinforcement Learning Model	Determines suitable adaptive learning actions
Large Language Model	Provides explanations, doubt solving, quizzes, assignments and feedback
Learning Analytics Data	Supports performance visualization and recommendation generation
Web Application	Provides the user interface for students and faculty/admin

4.2 Hardware / Software / Fabrication Implementation & Modern Tools Used

Software Implementation

The project is implemented as a **web-based intelligent learning platform** consisting of three integrated software modules.

Module 1 — Personalized Learning Engine

The first module implements student profile management and learning behavior analysis. Student interactions such as quiz scores, attempts, incorrect answers, topic completion and learning

consistency are used to construct the learner's current state. The Meta-Reinforcement Learning component then selects an appropriate action such as revision, additional practice, difficulty adjustment, or progression to a new topic.

Module 2 — AI Tutor Using Large Language Models

The second module provides an interactive AI tutoring environment. The student's current learning state and topic information are used to generate personalized explanations and responses. The module also supports adaptive quizzes, assignments, hints, examples, simplified explanations and instant feedback.

Module 3 — Performance Analytics & Recommendation Dashboard

The third module collects and processes learning performance information. It presents progress, topic mastery, learning trends, weak areas and improvement recommendations through dashboard-based visualizations. The resulting learning insights are returned to Module 1 to update the student's learning state.

Modern Tools Used

Tool / Software / Equipment	Purpose	Stage Used
Python	AI/ML logic, data processing and backend development	Development
Meta-Reinforcement Learning	Adaptive learning policy and action selection	Module 1
Large Language Model	AI tutoring and educational content generation	Module 2
HTML/CSS/JavaScript	Web interface development	Frontend Development
Database	Storage of student profiles, learning activities and performance	Development & Integration
Data Visualization Library	Performance graphs and learning analytics	Module 3
Git/GitHub	Version control and project management	Development

4.3 Test Plan & Verification

Testing was planned before evaluating the final implementation. Functional testing was used to verify whether each system requirement produced the expected behavior. Integration testing was performed to verify information flow between the three modules, while performance and usability checks were used to identify response-time and interface issues.

Requirement	Test Method	Specification Required Value	Acceptance Criterion	Achieved Value
Student profile creation	Create and save sample student profiles	Profile data stored correctly	Profile successfully created and retrieved	Pass
Learning behavior analysis	Submit different quiz results and activities	Learning state changes according to activity	Relevant behavior indicators updated	Pass
Adaptive recommendation	Provide different performance conditions	Recommendation should reflect learner state	Suitable next activity generated	Pass
Learning path	Simulate strong and weak topics	Path should prioritize learning needs	Personalized sequence generated	Pass
AI explanation	Ask questions at different knowledge levels	Context-appropriate explanation	Relevant explanation generated	Pass
Doubt solving	Submit student questions	AI should provide contextual responses	Response generated successfully	Pass
Adaptive quiz	Generate quizzes for different difficulty levels	Questions should reflect selected topic/level	Appropriate quiz generated	Pass
Assignment generation	Request assignment for weak topic	Assignment should target learning requirement	Personalized assignment generated	Pass
Instant feedback	Submit correct and incorrect responses	Feedback should identify response status	Feedback displayed	Pass
Progress analytics	Record learning activities	Dashboard should reflect updated data	Performance visualization updated	Pass
Weak-area detection	Provide low topic scores	Weak topics should be identified	Weak areas displayed	Pass
Recommendation feedback loop	Complete learning activity	Updated state should reach Module 1	Next recommendation changes accordingly	Pass
Faculty monitoring	Open faculty/admin dashboard	Student performance should be visible	Monitoring information displayed	Pass

4.4 Results

The implemented prototype demonstrates the integration of the three core modules into a continuous adaptive learning system. Functional testing confirmed the major workflows including student profile creation, learning behavior analysis, adaptive recommendation, AI tutoring, assessment generation, performance analytics and feedback-based learning-state updates.

Results Summary

Test Area	Result	Status
Student Profile	Profile created and retrieved successfully	Pass
Learning Behavior Analysis	Activity and performance information processed	Pass
Adaptive Learning Path	Learning sequence adjusted according to learner state	Pass
Dynamic Recommendation	Suitable next activities generated	Pass
AI Tutor	Personalized responses generated	Pass
Adaptive Quiz	Topic/difficulty-based questions generated	Pass
Assignment Generation	Personalized assignments generated	Pass
Instant Feedback	Response evaluation and explanation provided	Pass
Performance Dashboard	Learning progress and trends displayed	Pass
Weak Area Analysis	Weak topics identified from performance data	Pass
Module Integration	Data transferred across the three modules	Pass
Continuous Learning Loop	Updated learning state influenced subsequent recommendations	Pass

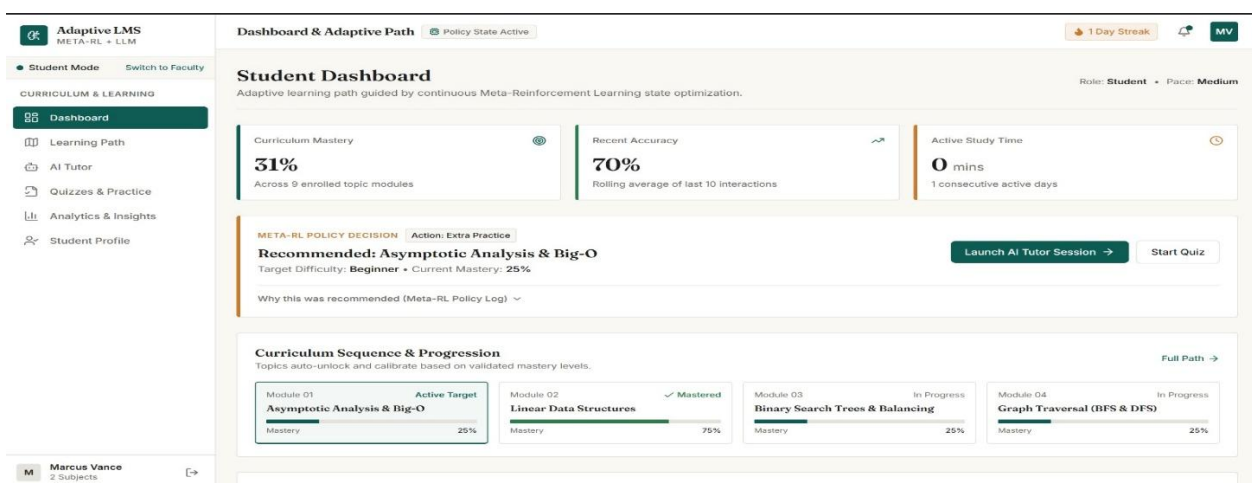


Figure 2: Student Dashboard and Adaptive Learning Recommendation

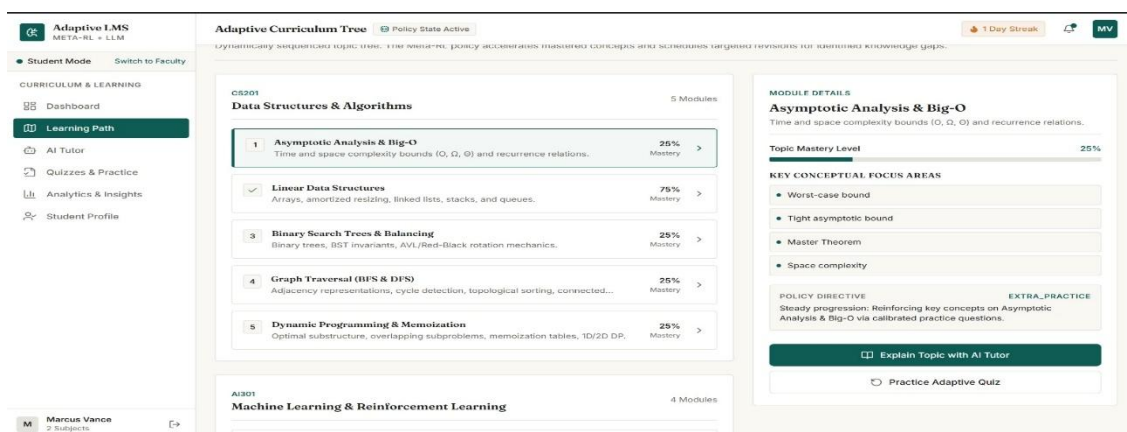


Figure 3: Adaptive Curriculum Learning Path

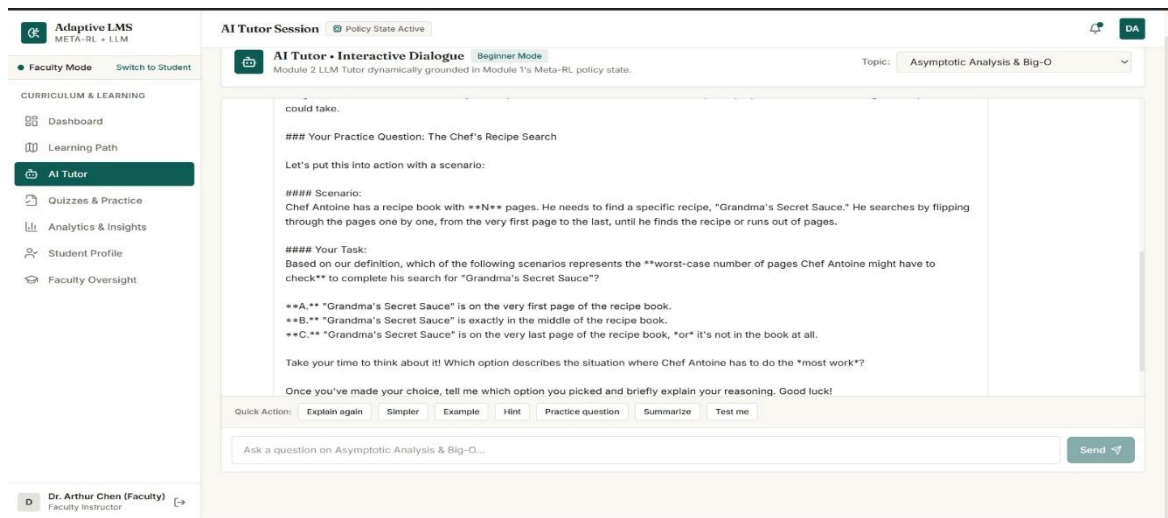


Figure 4: AI Tutor Interactive Learning Session

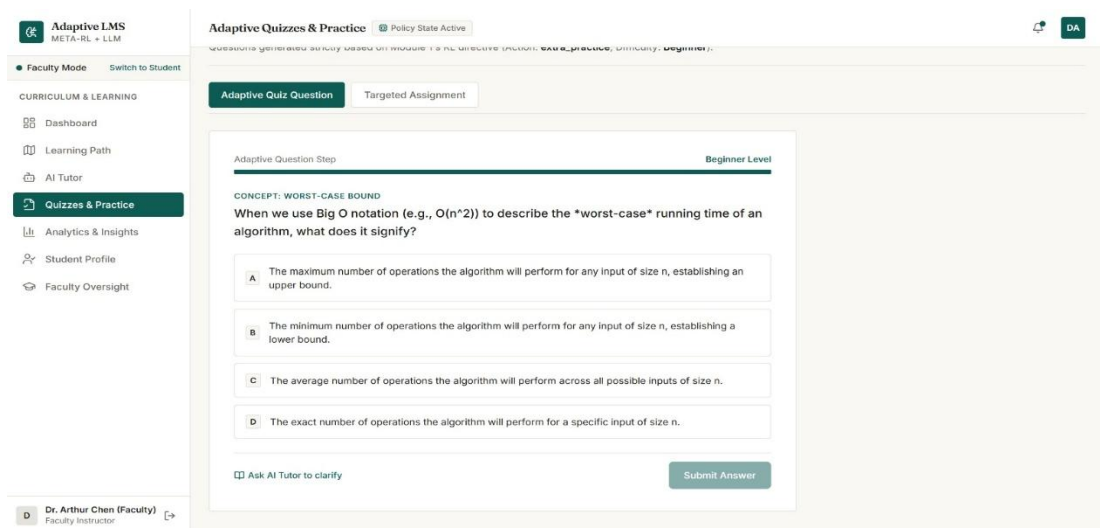


Figure 5: Adaptive Quiz and Practice Interface

4.5 Data Analysis & Engineering Judgment

The testing results indicate that the proposed system successfully connects student activity, adaptive decision-making, AI tutoring, and performance analytics. The learner state changes when new performance information is introduced, allowing the recommendation component to select different learning activities according to the student's needs.

Students demonstrating weaker performance can be directed toward revision or additional practice, whereas students demonstrating stronger performance can be progressed toward higher difficulty content. This behavior demonstrates the intended adaptive characteristic of the system.

The AI Tutor provides flexibility that is difficult to achieve through fixed educational content because

explanations and practice activities can be generated according to the current learning context. However, LLM-generated responses may not always be perfectly accurate, so AI output should be treated as assisted educational guidance rather than an unquestionable source.

The engineering evaluation therefore indicates that **integration is the main strength of the system**, while data quality, AI reliability, and computational requirements remain important factors for future improvement.

4.6 Comparison with Existing Solutions & Achievement of Objectives

Objective	Evidence	Achievement
Design a three-module adaptive learning architecture	Three integrated core modules implemented	Fully Achieved
Develop student profiles and learning behavior analysis	Student profile and activity tracking implemented	Fully Achieved
Model adaptive learning using Meta-RL	Learning state, actions and reward-based adaptation implemented	Fully Achieved
Implement personalized learning paths	Dynamic topic/activity recommendations implemented	Fully Achieved
Develop LLM-based AI Tutor	Explanation, doubt solving and learning assistance implemented	Fully Achieved
Generate adaptive quizzes and assignments	Dynamic assessment functionality implemented	Fully Achieved
Provide instant feedback	Student response evaluation and feedback implemented	Fully Achieved
Develop performance analytics	Progress, topic mastery and trend analysis implemented	Fully Achieved
Generate personalized recommendations	Analytics connected to adaptive recommendations	Fully Achieved
Integrate all three modules	Continuous learning feedback loop demonstrated	Fully Achieved
Test the complete system	Functional and integration testing performed	Fully Achieved
Evaluate long-term learner improvement	Prototype-level evaluation performed	Partially Achieved

The final objective is marked **Partially Achieved** because demonstrating long-term educational improvement would require a larger student population and longer-duration real-world evaluation.

4.7 Strengths & Limitations

Strengths

- Three clearly defined but interconnected intelligent learning modules.
- Personalized learning paths based on individual student behavior.
- Meta-Reinforcement Learning supports continuous adaptation.
- LLM-based tutor provides flexible and interactive learning assistance.
- Adaptive quizzes and assignments target individual learning needs.
- Analytics provide visual understanding of student performance.
- Weak areas and learning gaps can be identified automatically.
- Recommendations are connected back to the learning engine.
- Faculty/admin users can monitor student progress.
- Modular architecture supports future expansion.

Limitations

- The effectiveness of personalization depends on the quality and quantity of student data.
- LLM-generated content can occasionally contain inaccurate information.
- Meta-RL requires careful reward and state design.
- Large-scale real-world validation with many students is not covered by the prototype evaluation.
- AI-based processing may require additional computational resources as the number of users increases.
- Long-term learning improvement requires extended deployment and comparison with conventional learning approaches.

4.8 Project Planning, Roles & Budget

Project Milestone Timeline

Phase	Major Activity	Duration / Stage
Phase 1	Problem identification and literature review	Week 1

Phase	Major Activity	Duration / Stage
Phase 2	System requirements and architecture design	Week 2
Phase 3	Module 1 development	Weeks 3–4
Phase 4	Module 2 development	Weeks 5–6
Phase 5	Module 3 development	Weeks 7–8
Phase 6	Module integration	Week 9
Phase 7	Testing and debugging	Week 10
Phase 8	Results analysis and documentation	Week 11
Phase 9	Final verification and presentation preparation	Week 12

Roles / Contribution

Student	Assigned Responsibility	Actual Contribution
Chandana M. (192472250)	Module 1 – Personalized Learning Engine	Student profile, learning behavior analysis, adaptive learning path and Meta-RL implementation/testing
Yoga Madha A. (192425058)	Module 2 – AI Tutor Using Large Language Models	AI tutoring, personalized explanations, quizzes, assignments and feedback
Kushitha G. (192425329)	Module 3 – Performance Analytics & Recommendation Dashboard	Performance analytics, visualizations, recommendations and dashboard testing

CHAPTER 5

CONCLUSION, FUTURE WORK AND REFLECTION

5.1 Conclusion

The project “**Personalized Adaptive Learning Management System Using Meta-Reinforcement Learning and Large Language Models**” was developed to address the limitations of conventional learning platforms that provide similar learning experiences to students with different abilities, learning speeds, and performance levels. The proposed system provides an intelligent and personalized approach by continuously analyzing student learning behavior and adapting the learning experience.

The implementation and testing demonstrated that the three modules can operate as a continuous adaptive learning cycle. Student activity is captured, analyzed, used to update the learning state, and subsequently influences future learning recommendations. Functional and integration testing verified the major system operations, including profile management, adaptive recommendations, AI tutoring, assessment generation, performance analysis, and information flow between modules.

Overall, the project demonstrates the feasibility of combining **Meta-Reinforcement Learning and Large Language Models** within a unified adaptive learning platform. The resulting system provides a foundation for delivering more individualized, interactive, and data-driven learning experiences.

5.2 Future Work

The proposed system can be further enhanced in several ways:

- **Large-Scale Student Evaluation:** Test the system with a larger number of students over a longer period to measure actual learning improvement.
- **Advanced Meta-RL Models:** Explore more sophisticated Meta-Reinforcement Learning algorithms for improved personalization and faster adaptation.
- **Improved Learner Modelling:** Incorporate additional behavioral indicators such as engagement patterns, response time, learning consistency, and preferred learning formats.
- **Knowledge Graph Integration:** Connect concepts and prerequisites through knowledge graphs to improve adaptive topic sequencing.

- **Multimodal AI Tutor:** Extend the AI Tutor to support images, diagrams, voice interaction, and educational videos.
- **Improved AI Reliability:** Introduce stronger response validation and educational-content verification mechanisms to reduce inaccurate LLM-generated responses.
- **Explainable Recommendations:** Provide clearer explanations of why a particular topic, quiz, or learning activity was recommended.
- **Mobile Application:** Develop Android and iOS applications to provide learning access across different devices.
- **Advanced Faculty Analytics:** Add early-warning indicators to help faculty identify students who may require academic support.
- **Scalable Deployment:** Deploy the system on cloud infrastructure to support larger numbers of concurrent users.
- **Privacy Enhancement:** Implement stronger privacy-preserving mechanisms for storing and processing student learning data.
- **Long-Term Adaptive Evaluation:** Compare the proposed system with conventional learning approaches using measurable learning-outcome indicators.

5.3 Professional Learning & Individual Reflection

New Knowledge Acquired

- Learned how **Meta-Reinforcement Learning** can be applied to personalized learning and sequential decision-making.
- Developed an understanding of **Large Language Models** and their application in AI-based tutoring.
- Learned how learner behavior can be represented as a changing learning state.
- Gained knowledge of adaptive learning paths and dynamic content recommendation.
- Learned how learning analytics can convert student activity data into meaningful educational insights.

- Understood the importance of integrating multiple AI components into a continuous feedback loop.

Engineering & Professional Skills Developed

- System architecture and modular software design.
- AI/ML model development and integration.
- LLM-based application development.
- Data analysis and performance visualization.
- Web application development.
- Functional and integration testing.
- Debugging and problem solving.
- Git-based version control and collaborative development.
- Technical documentation and engineering report preparation.
- Teamwork, task allocation and project planning.
- Engineering decision-making based on system requirements and testing evidence.

Individual Reflection — Yoga Madha A. (192425058)

The most difficult engineering problem was integrating the **Large Language Model-based AI Tutor** with the student's learning context while maintaining personalized responses. This was addressed by connecting tutoring interactions with the learner's current topic, performance, and learning state. A key engineering decision was to provide different forms of assistance such as explanations, examples, hints, quizzes, and feedback instead of using the tutor only for question answering. I independently developed knowledge of LLM-based educational applications and prompt-based interaction. If the project were repeated, I would add stronger response validation and multimodal tutoring capabilities.

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APPENDICES

APPENDIX A — DETAILED CALCULATIONS

A.1 Learner State Representation

The student's learning state is represented using measurable learning indicators such as:

- Current knowledge level
- Quiz accuracy
- Number of attempts
- Incorrect answers
- Topic completion
- Learning time
- Recent improvement
- Learning consistency

A simplified learner-state representation can be expressed as:

Learner State =

[Knowledge Level,

Quiz Accuracy,

Attempts,

Learning Time,

Topic Completion,

Mistake Rate,

Improvement,

Consistency]

Each parameter can be normalized to a common scale to make the values suitable for adaptive decision-making.

A.2 Reward Calculation

The Meta-Reinforcement Learning component uses learning outcomes to calculate a reward.

Reward =

Performance Improvement

+ Engagement

+ Accuracy

+ Completion

– Mistake Penalty

The reward is used as feedback to determine whether the selected learning action was beneficial.

A.3 Topic Mastery

Topic mastery can be estimated using performance indicators:

Topic Mastery =

Weighted Performance

+ Completion

+ Recent Improvement

– Repeated Mistakes

The resulting value can be mapped into categories such as:

Score Range	Mastery Level
0–39	Beginner
40–59	Developing
60–79	Proficient
80–100	Mastered

A.4 Adaptive Action Selection

Possible actions considered by the learning engine are:

$A = \{$

Introduce New Topic,

Revise Topic,

Increase Difficulty,

Decrease Difficulty,

Additional Practice,

Alternative Resource

}

The action with the highest estimated suitability for the current learner state is selected.

APPENDIX B — ENGINEERING DRAWINGS / SCHEMATICS

B.1 Overall System Architecture

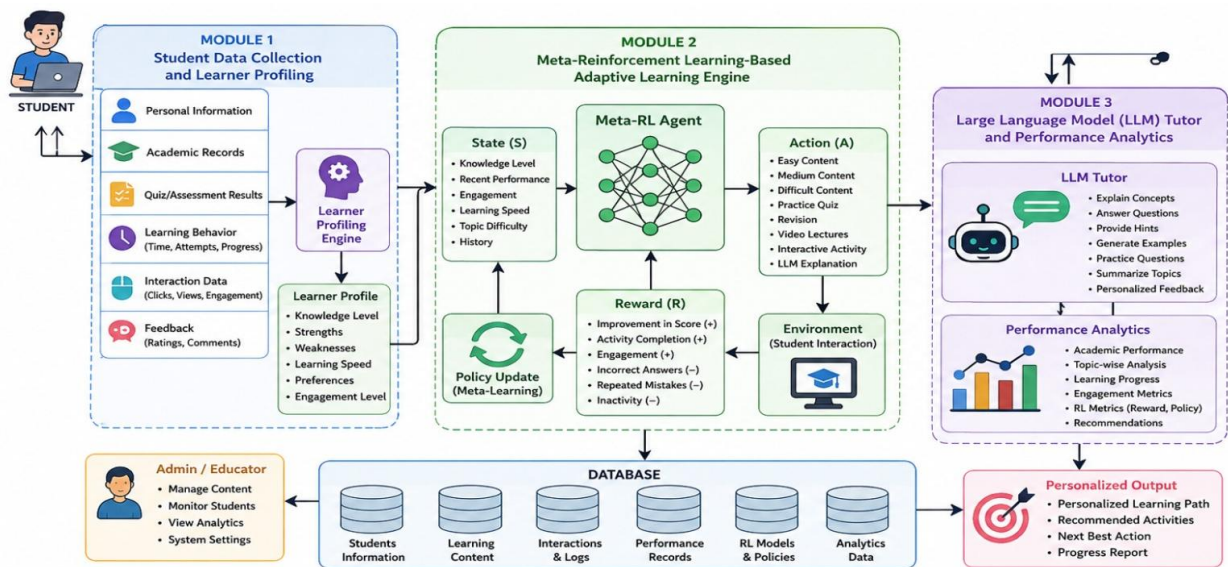
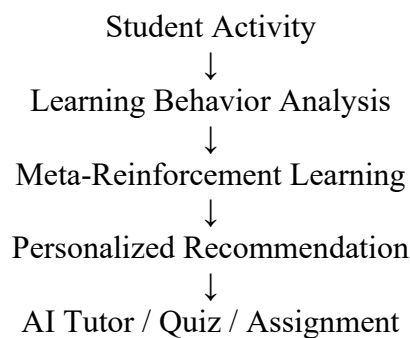
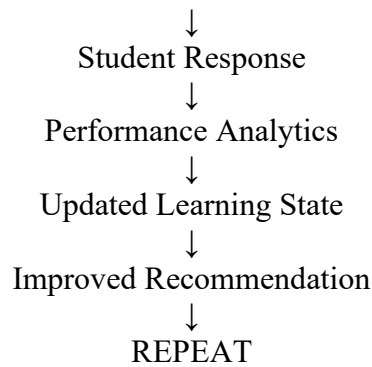


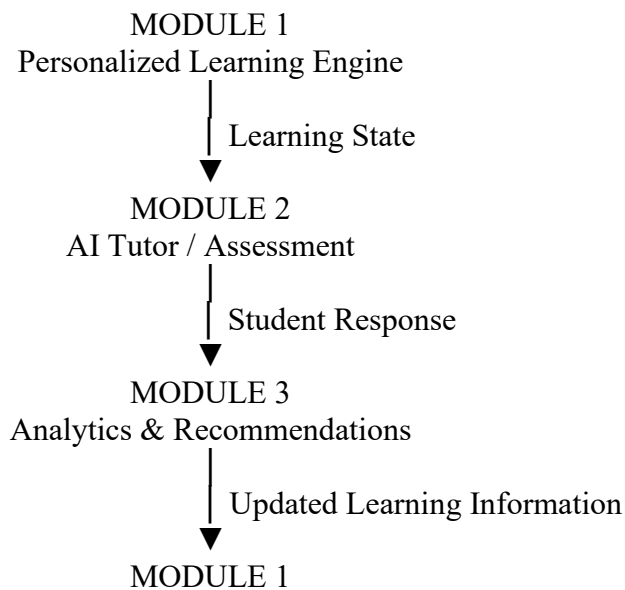
Figure 1: System Architecture

B.2 Continuous Adaptive Learning Cycle





B.3 Module Interaction

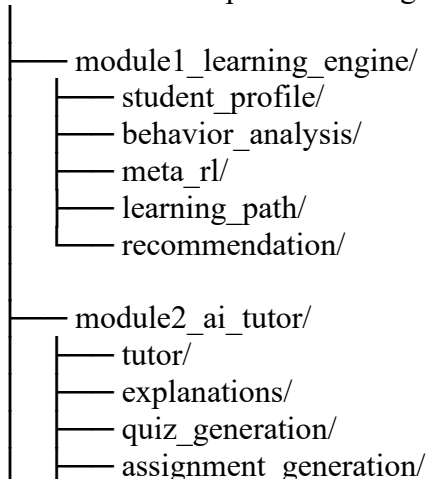


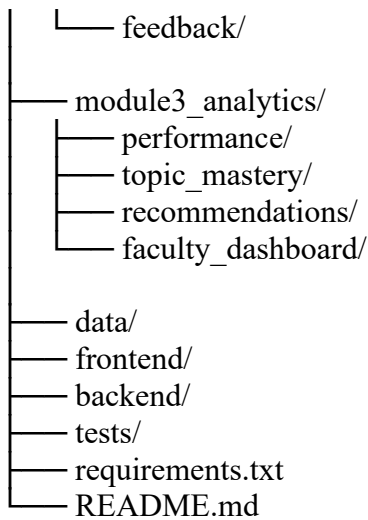
APPENDIX C — SOURCE CODE / CODE REPOSITORY INFORMATION

The complete source code should be maintained separately in the approved project repository. Only essential code excerpts should be included in the main report.

C.1 Recommended Repository Structure

Personalized-Adaptive-Learning/





C.2 Essential Code Excerpts

The report may include short excerpts showing:

- Learner-state construction
- Reward calculation
- Adaptive action selection
- AI Tutor request handling
- Quiz generation
- Performance calculation
- Recommendation generation
- Module-to-module data flow

The complete implementation should be referenced through the team's approved repository rather than reproducing every source file in the report.

APPENDIX D — DATASHEETS

This software project does not require conventional electronic hardware datasheets.

The following **software/AI technical documentation** may instead be attached where applicable:

Resource	Purpose
Meta-RL algorithm documentation	Adaptive policy implementation

Resource	Purpose
LLM/API documentation	AI Tutor integration
Python documentation	Software implementation
Database documentation	Student and learning-data management
Visualization library documentation	Analytics dashboard development

APPENDIX E — EXPERIMENTAL / RAW DATA

The following data should be maintained as the experimental dataset for system testing.

E.1 Sample Student Learning Data

Student	Topic	Quiz Score (%)	Attempts	Learning Time (min)	Completion (%)	Mastery
S001	Mathematics	82	1	35	100	Mastered
S001	Algebra	58	3	48	70	Developing
S002	Mathematics	91	1	28	100	Mastered
S002	Algebra	74	2	39	90	Proficient
S003	Physics	46	4	55	60	Developing
S003	Mechanics	38	4	62	50	Beginner

E.2 Adaptive Recommendation Example

Student	Observed Condition	Selected Action	Reason
S001	High mastery	Increase difficulty	Student demonstrates strong performance
S001	Low performance Algebra	Additional practice	Repeated mistakes detected
S002	High topic accuracy	New topic	Current topic demonstrates sufficient mastery
S003	Low Physics score	Revise topic	Learning gap detected
S003	Repeated mistakes	AI Tutor explanation	Concept clarification required

APPENDIX F — ETHICS / INSTITUTIONAL APPROVAL

The project primarily uses educational learning information for system development and evaluation.

Where real student data is collected, the following ethical measures should be followed:

- Obtain appropriate institutional approval where required.
- Inform participants about the purpose of data collection.
- Collect only information necessary for the learning system.

- Avoid unnecessary personally identifiable information.
- Restrict access to student learning records.
- Use anonymized or pseudonymized data for experiments wherever possible.
- Allow participants to understand how their learning information is used.
- Do not use AI recommendations as a replacement for professional academic judgment.
- Clearly identify AI-assisted educational interactions.

APPENDIX G — PROJECT MEETING / PROGRESS RECORDS

Meeting / Date	Discussion / Activity	Decision / Outcome
Meeting 1	Project problem and objectives discussed	Personalized adaptive learning system selected
Meeting 2	System architecture discussed	Three-module architecture finalized
Meeting 3	Module 1 design discussed	Meta-RL selected for adaptive learning
Meeting 4	Module 2 development discussed	LLM-based AI Tutor planned
Meeting 5	Module 3 design discussed	Analytics and recommendation dashboard planned
Meeting 6	Module integration reviewed	Data flow between modules established
Meeting 7	Testing discussed	Functional and integration test cases prepared
Meeting 8	Final results reviewed	Improvements and documentation completed

APPENDIX H — USER / INDUSTRY FEEDBACK

Where feedback is collected from students, faculty, or industry reviewers, it can be documented using the following format.

Reviewer Type	Feedback	Action Taken
Student	Personalized recommendations were useful	Recommendation interface refined
Student	AI explanations should be simpler when concepts are difficult	Simplified explanation option added
Faculty	Weak-topic identification is useful for monitoring	Topic analytics included
Faculty	Performance trends should be visible over time	Trend visualization included
Reviewer	AI output should be treated carefully	AI response limitations documented

APPENDIX I — PUBLICATIONS / PATENTS / COMPETITIONS / PRODUCT OUTCOMES

I.1 Publications

No research publication was produced as part of the current prototype unless otherwise specified.

Publication: [Insert publication details if applicable]

I.2 Patents

No patent was filed for the current prototype.

Patent Reference: [Insert patent number if applicable]

I.3 Competitions

Competition / Hackathon: [Insert name if applicable]

Achievement: [Insert achievement if applicable]

I.4 Product Outcome

The primary project outcome is a **prototype Personalized Adaptive Learning Management System** integrating:

- Meta-Reinforcement Learning
- Large Language Models
- Adaptive learning paths
- AI tutoring
- Adaptive assessments
- Performance analytics
- Personalized recommendations

APPENDIX J — INDIVIDUAL CONTRIBUTION EVIDENCE

J.1 Individual Evidence — Yoga Madha A

Module: Module 2 – AI Tutor Using Large Language Models

Evidence to attach:

- AI Tutor interface screenshot
- Personalized explanation example
- Doubt-solving interaction
- Adaptive quiz screenshot
- Assignment generation example
- Instant feedback example
- Module 2 testing results
- Relevant Git commits/source-code contribution